



Department of Artificial Intelligence and Data Science

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: B.Tech , V & Artificial Intelligence and Data Science

Course Code & Title: AD3002 & Health Care Analytics

Name of the Faculty member (s): Dr.R.Ramana

Innovative Practice Description

- **Unit / Topic:** Unit II / Analytics on Machine Learning

- **Course Outcome:** CO2

- **Topic Learning Outcome:** TLO7

- **Activity Chosen:** Think–Pair–Share

- **Justification:**

A Think–Pair–Share activity was conducted to enhance students' understanding of Python Variables and Types, Data Structures and Containers, and Pandas. This activity encouraged active participation and peer learning, helping students apply Python concepts to healthcare datasets.

- **Time Allotted for the Activity:** 45 Minutes

- **Details of the Implementation:**

- ❖ The **Think–Pair–Share (TPS)** method was implemented in three stages:

- ❖ **Stage 1: THINK (Individual Work)**

1. Each student was given short tasks/questions to think independently:

- ❖ Identify variable types used in patient data (int, float, string, boolean).
- ❖ Explain the difference between lists, tuples, sets, and dictionaries in Python.
- ❖ Write a simple Python code snippet using Pandas to create a DataFrame for patient records.

- ❖ **Stage 2: PAIR (Collaborative Work)**

Students paired up to discuss their answers.

- ❖ Each pair compared their Python code examples and discussed efficiency in representing healthcare data (e.g., storing patient details, lab test results).
- ❖ Pairs explored how Pandas simplifies healthcare data manipulation and cleaning.

- ❖ **Stage 3: SHARE (Class Discussion)**

Selected pairs presented their solutions to the class.

The instructor facilitated discussion on:

- ❖ Common data types used in healthcare datasets.
- ❖ Advantages of Pandas for handling electronic health records (EHRs).
- ❖ Real-world healthcare data cleaning examples using Pandas functions like `dropna()`.



- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students found the activity engaging as it helped them visualize how Python can be used to handle patient data efficiently.
- They appreciated the peer discussion phase, which clarified syntax doubts and improved coding confidence.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- The activity encouraged active participation from all students.
- Peer discussion enhanced understanding of data structures and Pandas operations.
- Students demonstrated improved problem-solving and analytical thinking relevant to healthcare datasets.

- ❖ **Challenges faced in implementation:**

- Some students initially struggled to differentiate between Python data structures such as lists, tuples, and dictionaries.
- Limited prior coding experience made it difficult for a few students to write and debug Pandas programs.
- Time constraints during the sharing phase reduced opportunities for all pairs to present their solutions.
- Internet connectivity issues affected students using online Python environments like Google Colab.
- Ensuring equal participation from all members in pairs required additional instructor monitoring.



FEEDBACK

Active Learning Best practices: Think-Pair-Share

Degree, Semester & Branch : V Semester & B. Tech AI / DS
Course Code & Title : AD3002 & Health Care Analytics
Name of the Faculty member (s) : Dr. R. Ramana, AP / AD

Theme of discussion: Python Variables and Types, Data Structures and Containers, Pandas

Date and Time: 25.08.2025 & 10.30 am to 11.15 am

Feedback collected in class and also through online

Feedback Questionnaire:

1. Does the activity encourage active learning? **Yes No**
2. Do you have a clear understanding about Python Variables and Types, Data Structures and Containers, Pandas?
Excellent Very Good Good Satisfactory
3. Does this activity learning improve listening, communication and problem-solving skills?
Excellent Very Good Good Satisfactory

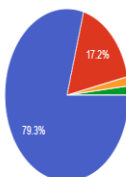
Feedback Analysis:

1. Does the activity encourage active learning?
58 responses



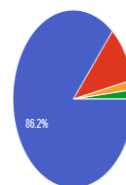
● YES
● NO

2. Do you have a clear understanding about Python Variables and Types, Data Structures and Containers, Pandas?
58 responses



● Excellent
● Very Good
● Good
● Satisfactory

3. Does this activity learning improve listening, communication and problem-solving skills?
58 responses



● Excellent
● Very Good
● Good
● Satisfactory

Faculty in-charge

HOD/AD